THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 10

Full Syllabus (Chapters 1–8) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 17 units	$\sqrt{64+225}=17$
2	**(d)** 13 units	$\sqrt{(-7-(5))^2+(6-(1))^2}=\sqrt{144+25}=13$
3	**(b)** on the y-axis	$x=0$, so the point lies on the y-axis
4	**(b)** 10 cm	$4x=10$
5	**(b)** -11	$-2(5)+(-1)=-11$
6	**(d)** irrational	not a ratio of integers
7	**(c)** 14	Brahmagupta's rules for signs
8	**(a)** $\frac{161}{450}$	standard method ⇒ $\frac{161}{450}$
9	**(b)** $9s^2-25t^2$	$(a+b)(a-b)=a^2-b^2$
10	**(c)** $16x^2+24x+9$	$(a+b)^2=a^2+2ab+b^2$
11	**(c)** longer	longer chord ⇔ closer to the centre
12	**(d)** 65°	angle at the centre is double the angle on the circle
13	**(a)** 82°	central angle is double
14	**(b)** 36 cm^2	$s=18$, area $=\sqrt{18\cdot9\cdot8\cdot1}=36$
15	**(b)** 7 cm	$r=\frac{C}{2\pi}=\frac{44\times7}{44}=7$
16	**(c)** 22 cm	$2\pi r=2\times\frac{22}{7}\times3.5=22$
17	**(c)** 1	always happens
18	**(b)** 120	expected number $=P\times$ population
19	**(a)** neither an AP nor a GP	differences 2,3,4,5 are not constant
20	**(b)** 96	$ar^{n-1}=3\times(2)^5=96$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	Midpoint $=\left(\frac{4+(10)}{2},\frac{6+(-2)}{2}\right)=(\mathbf{7},\ \mathbf{2})$	1+1
22	$a_n=5n+(-3)$; $a_{12}=5(12)+(-3)=\mathbf{57}$	1+1
23	$4-15=-11$ °C.	1+1
24	$(4y-3)^2$	2
25	$\angle A+\angle C=180^\circ$ and $\angle B+\angle D=180^\circ$, so yes — opposite angles add up to 180° and the quadrilateral is cyclic.	1+1
26	$2\times\frac{22}{7}\times10=62.857\ldots\approx62.9$ cm.	1+1
27	Experimental $=\frac{8}{50}=0.16$; theoretical $=\frac{1}{6}\approx0.167$. They are close.	1+1
28	$-3,\ -8,\ -13,\ -18,\ -23$.	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Distances: $\sqrt{16+16}=4\sqrt{2}$ from each. The three points form an isosceles triangle with the apex at $(0,-1)$; base $=8$ and $64=32+32$, so it is also right-angled at the apex.	1+1+1
30	For $x=0,1,2$: $(0,-2)$, $(1,1)$, $(2,4)$. It cuts the y-axis at $(0,-2)$.	2+1
31	Take $OA=1$; draw a perpendicular at A and mark $AB=1$. By Baudhāyana–Pythagoras $OB=\sqrt{2}$. With centre O and radius OB, draw an arc to cut the number line at P; P represents $\sqrt{2}$.	1+2
32	$(n+1)^2-n^2=2n+1=n+(n+1)$ ✓.	1+1+1
33	Area $=20\times9=180$ cm^2 (1+1). Using the 15 cm side as base: $15\times h=180\Rightarrow h=12$ cm (1).	2+1
34	(i) 5, 10, 15, 20, 25 ⇒ $\frac{5}{25}=\frac{1}{5}$ (ii) 2,3,5,7,11,13,17,19,23 ⇒ $\frac{9}{25}$.	1+2
35	Half-chord $=\sqrt{r^2-d^2}$ (Baudhāyana–Pythagoras) (1); chord $=2\sqrt{r^2-d^2}$ (1). For $r=10, d=6$: $2\sqrt{64}=16$ (1).	1+1+1

	OR: $\angle BDC = \angle BAC = 30^\circ$ (angles in the same segment) (1+1). $\angle CBD = 180^\circ - 100^\circ - 30^\circ = 50^\circ$ (1).	2+1
36	$t_n = \frac{n(n+1)}{2}$: $t_{10} = 55$, $t_{17} = 153$, $t_{80} = 3240$.	1+1+1
	OR: $1, 2, 3, 5, 8, 13, 21, 34, 55, 89$.	3

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $\frac{1}{4} - \frac{5}{12} = \frac{3-5}{12} = -\frac{1}{6}$; $-\frac{1}{6} \div \frac{2}{3} = -\frac{1}{6} \times \frac{3}{2} = -\frac{1}{4}$. (ii) Sum $= \frac{6}{10} + \frac{7}{10} = \frac{13}{10}$; product $= \frac{21}{50}$.	3+2
	OR: (i) $\frac{x}{2} = \frac{9}{2} \Rightarrow x = 9$. (ii) LHS $= \frac{1}{2} \times \frac{13}{12} = \frac{13}{24}$; RHS $= \frac{3}{8} + \frac{1}{6} = \frac{9}{24} + \frac{4}{24} = \frac{13}{24}$ ✓	2+3
38	(i) Diagonal $= \sqrt{25 + 144} = 13$; angle in a semicircle ⇒ it is a diameter, radius 6.5 (2). (ii) $2 \times \frac{1}{2} \times 5 \times 12 = 60$ sq. units (2). (iii) Its sides in order are 5,5,12,12 (a kite) — opposite sides are not equal, so it is not a rectangle (1).	2+2+1
	OR: (i) 80° (1). (ii) $OB = OC$ ⇒ $\angle OBC = 50^\circ$ (1). (iii) Same segment ⇒ $\angle BDC = 40^\circ$ (1). (iv) Half-chord $= \sqrt{60^2 - 36^2} = 48$; $BC = 96$ cm (2).	1+1+1+2
39	Inscribed hexagon: six equilateral triangles of side r ⇒ perimeter $6r < 2\pi r$ ⇒ $\pi > 3$ (2). Circumscribed hexagon: side $= \frac{2r}{\sqrt{3}}$ ⇒ perimeter $4\sqrt{3}r > 2\pi r$ ⇒ $\pi < 2\sqrt{3} \approx 3.46$ (2). More sides squeeze the two perimeters closer, trapping π more tightly (1).	2+2+1
	OR: (i) $1 \times 19 = 19$, $2 \times 18 = 36$, $3 \times 17 = 51$, $4 \times 16 = 64$, $5 \times 15 = 75$, $6 \times 14 = 84$, $7 \times 13 = 91$, $8 \times 12 = 96$, $9 \times 11 = 99$, $10 \times 10 = 100$ (3). (ii) Greatest 100 (10×10), least 19 (1×19) (1). (iii) It is a square (1).	3+1+1
40	(i) 100, 150, 200 (1). (ii) $t_n = 100 + 50(n-1) = 50n + 50$ (1). (iii) $t_{12} = 650$; total $= \frac{12}{2}(100 + 650) = ₹4,500$ (2). (iv) $50n + 50 = 1000 \Rightarrow n = 19$ (1).	1+1+2+1
	OR: (i) $-5, -2, 1, 4, 7$ (1). (ii) $t_n = 3n - 8$ (1). (iii) $3n - 8 = 52 \Rightarrow n = 20$, yes (1). (iv) $\frac{20}{2}(-5 + 52) = 470$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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